

Dev Garg

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EDUCATION

Texas A&M University

Master of Computer Science

- GPA: **4.0/4.0**
- Courses: Large Language Models, Software Engineering, Deep Learning

Present

College Station, TX

Indian Institute of Technology, BHU

Bachelor of Technology - Electronics Engineering

- GPA: 8.87/10.0

2017-2021

Varanasi, India

EXPERIENCE

Societe Generale GSC

Software Engineer

June 2021 - Dec 2023

Bengaluru, India

- Played an integral role in every phase of the Agile software development lifecycle for credit risk exposure calculation applications employing **Big Data** technologies.
- Achieved an **50%** reduction in validation time by implementing automatic regression tests. Additionally, designed an auto-scaling system slashing compute costs by approximately **€XX,XXX** per quarter. Led the development of multiple alerting features and provided mentorship to new team members.
- Exposure: *Java, Spring Boot, Apache Spark, Kafka, Elasticsearch, Jenkins, SQL, Azure*

Societe Generale GSC

Software Engineer Intern

May 2020 - June 2020

Remote

- Developed an **ML-based** incident resolution recommendation system, reducing operational risks and improving response times by **38%**. Worked closely with cross-functional teams to design, develop, and test the MVP, meeting strict deadlines.
- Exposure: *Python, Scikit-learn, NLTK, Pandas, spaCy, Networkx*

PROJECTS

Attention-based Model Architecture for Citation Graph

- Developed a novel reference-enhanced NLU model leveraging cited papers to generate context-aware and comprehensive research abstracts, integrating **LoRA fine-tuned LLaMA** for improved document representation.
- Improved *perplexity* from **14.1 to 6.8** for causal modeling and competitive in summarization task, demonstrating higher coherence with potential applications in academic and publishing tools.

Agentic Self-Corrective Retrieval-Augmented Generation

- Designed and implemented an adaptive **RAG** system leveraging *LangChain* and *ChromaDB* vector store based on user-supplied documents and URLs. Leveraged *Google Custom Search API* to complement document-based retrieval reducing hallucinations and increasing the robustness of generated answers.
- Evaluated system performance with *RAGAS* metrics, resulting in a **20%** increase in answer relevance and a **5%** enhancement in faithfulness. Deployed on *AWS EC2* using *Docker Compose* with an intuitive *Streamlit* UI, and a 4-bit quantized *LLaMA3* model operational via an *Ollama* server for seamless and scalable operations.

News Aggregation and Recommendation System

- Developed a web-based News Aggregator that periodically fetches RSS data from various news outlets, centralizing the latest stories. Implemented a **Hybrid Recommendation System** to suggest articles based on similarity, user behavior, news freshness, and personalized content curation according to user interests and reading history.
- Utilized *BERT* embeddings to capture semantic and contextual nuances and *pandas* for data manipulation. Deployed on *AWS Lambda* and *DynamoDB* for scalable serverless services, with an interface using *FAST API*.

TECHNICAL SKILLS

Languages: Python, C++, Java, Javascript, SQL

Frameworks: Tensorflow, Pytorch, React, Spring Boot, Spark

Tools/Platforms: Git, Azure Datalake, Docker, Kubernetes, AWS

Interests and Experience: Machine Learning, Big Data Systems, Web Development

Certifications: Google Cloud Professional MLE, Machine learning Spec., MLOps Spec., GenAI with LLMs